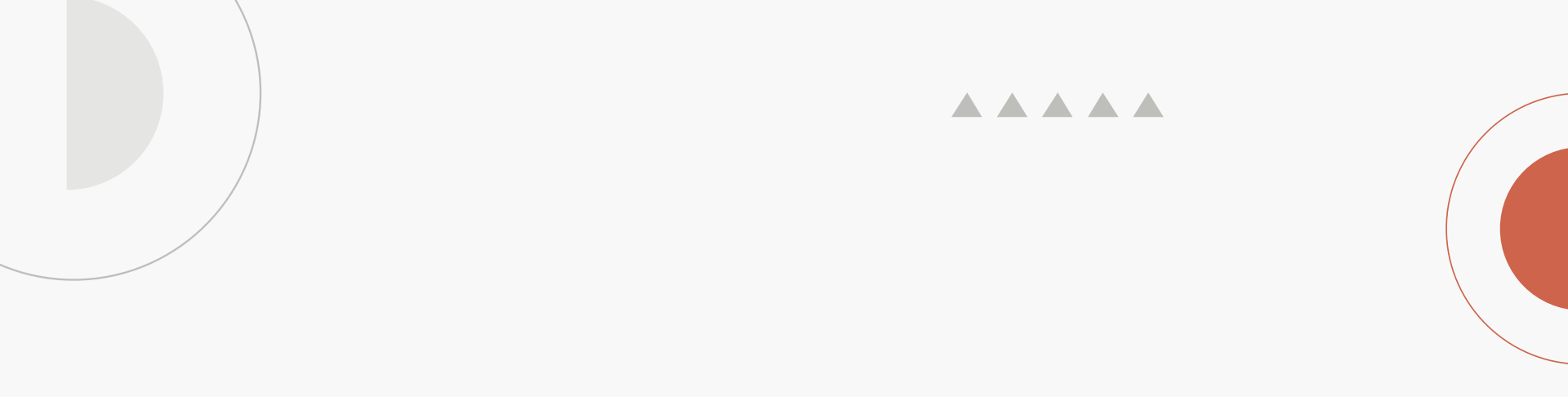
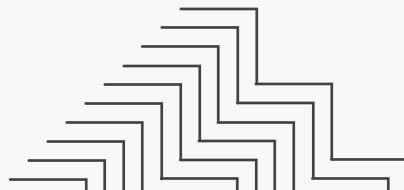




OOP-Project

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01

Part 1

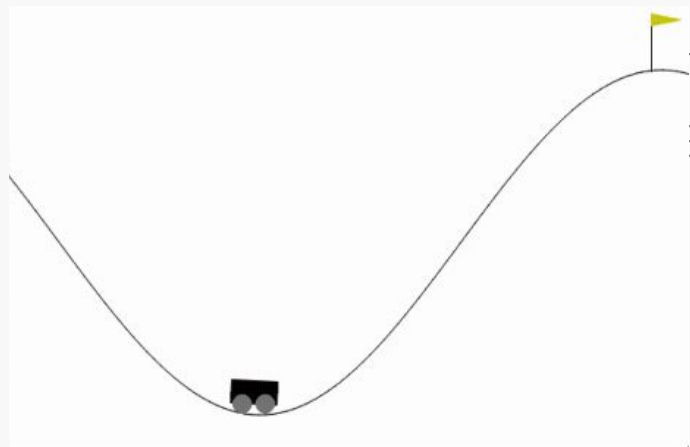


Mountain Car

We utilize Q-learning with the following equation:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \cdot \left[R_{t+1} + \gamma \cdot \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right]$$

Where the α is dynamically determine by the max learning rate and the decay of the current episode





02

Part 2



Frozen Lake

1. Reward Shaping w/ DistanceRewardWrapper
 - 到終點: +10
 - 掉洞: -10
 - 沒終止: 用「距離變化」當 reward: $\text{old_dist} - \text{new_dist}$
 - 走近終點 → 正 reward
 - 遠離終點 → 負 reward

→ Dense Feedback

2. $\text{min_exploration_rate} = 0.04$
3. Q-Learning (TD error) $\gamma = 0.95$





03

Part 3 – The Design



The Design

WareHouse Robot

This Map will generate random numbers of boxes between 3 to 8, then the robot will need to find the path to collect all the box and return to the “starting” green point

The Design



The Design: File structure

Obstacles:

Generate the Map with obstacles

Rewards:

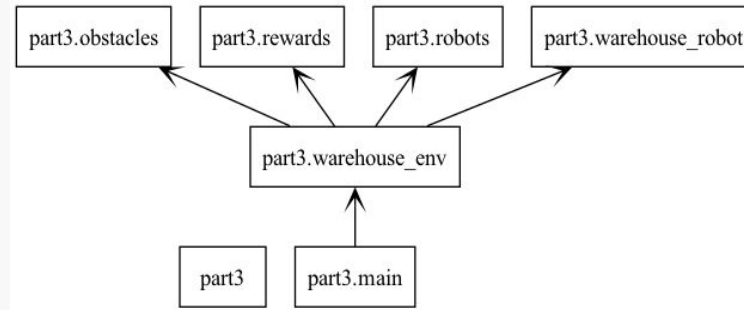
Decide the reward of each move

Robots:

A predefined robot rule-set

Warehouse_robot:

A trained agent that will load the model from the file







03

Part 3 - DEMO



DEMO

The Design



Git Actions

We utilize Git Action Workflow to auto evaluate the model result and auto PR back to the repo (And always replace the latest result)

The screenshot shows a GitHub Pull Request (PR) interface for a repository named 'TWCKaijin / OOP-project'. The PR is titled 'Agent Evaluation Results #1' and is created by 'github-actions'. It shows a merge of 1 commit from the 'agent-evaluation-results' branch into the 'main' branch. The PR is currently open and has 12 additions and 0 deletions. The PR description includes an 'Agent Evaluation Report' with the following details:

- Mountain Car Status**
 - Result: Final Mean Reward (Last 100 episodes): -183.97
- Frozen Lake Status**
 - Result: ✔ Success Rate: 74.00% (74 / 100 episodes)
- Warehouse Robot Status**
 - Result: Evaluation completed. Success Rate: 96.00%
 - Metrics: Avg Reward: 18.16

The right sidebar shows the 'Reviewers' section with 'No reviews' and the 'Assignees' section with 'No one—assign yourself'. The 'Labels' section shows 'None yet'. The 'Projects' section shows 'None yet'. The 'Milestone' section shows 'No milestone'. The 'Development' section is also visible.